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What Quantum Trace Can a Heavy Particle Leave?

Curvature–Spectator Entanglement
and the Cosmological Collider

Step 6: Heavy Clock and In-in Benchmark

Group-B

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1 Heavy-field clock and perturbative expectation value

Outside the horizon, discard x^2 terms. Integrating the light equation gives $\dot{Q} + g\Sigma \propto e^{-3N}$. For the decaying heavy branch, the transient can be neglected, so $\dot{Q} \simeq -g\Sigma$. The spectator obeys

$$\ddot{\Sigma} + 3\dot{\Sigma} + (M^2 + g^2)\Sigma = 0. \tag{1}$$

Substitution of $\Sigma \propto x^p = e^{-pN}$ yields $p^2 - 3p + M^2 + g^2 = 0$. Thus

$$p = \frac{3}{2} \pm i\mu, \qquad \mu = \sqrt{M^2 + g^2 - \frac{9}{4}}. \tag{2}$$

At $M = 2$, $g = 0.1$, $\mu = 1.3266499$. A negative square-root argument gives real powers, not an oscillating clock. The interaction-picture expansion, treating the mixed quadratic model exactly, gives

$$\Delta\langle O(\eta_f)\rangle = -i \int_{-\infty}^{\eta_f} \mathrm{d}\eta \, \langle [O_I(\eta_f), H_I(\eta)]\rangle_0. \tag{3}$$

For the test Hamiltonian $H_I = \gamma \int \mathrm{d}^3x \, a(q')^3/3!$, the six contractions cancel 3!. The massless derivative mode is $u_k' = Hk^2\eta e^{-ik\eta}/\sqrt{2k^3}$. The three late external modes and the factor $a = -1/(H\eta)$ reduce the time dependence to $\eta^2 e^{iK\eta}$, where $K = k_1 + k_2 + k_3$.

2 Analytic integral and numerical benchmark

With the vacuum regulator,

$$I_\epsilon = \int_{-\infty}^{0} \eta^2 e^{(\epsilon+i)K\eta} \mathrm{d}\eta = \frac{2}{[(\epsilon+i)K]^3}. \tag{4}$$

The commutator selects its imaginary part. Keeping the sign from a and the contractions gives

$$B_{\text{test}} = -\frac{\gamma H^5}{4k_1k_2k_3} \operatorname{Im} I_\epsilon \longrightarrow -\frac{\gamma H^5}{2k_1k_2k_3K^3}. \tag{5}$$

We check this independently with oscillatory sine quadrature, rather than using the analytic integral as the numerical answer. The tests use $\gamma = H = 1$ and $(k_1, k_2, k_3) = (r, 1, 1)$ with $r = 1, 0.1, 0.03, 0.01$. At $\epsilon = 10^{-3}$ the maximum relative error from the zero-regulator limit is 6.000×10^{-6}, comfortably below the required 0.1%. Agreement with the regulated analytic formula is also checked before taking the limit.

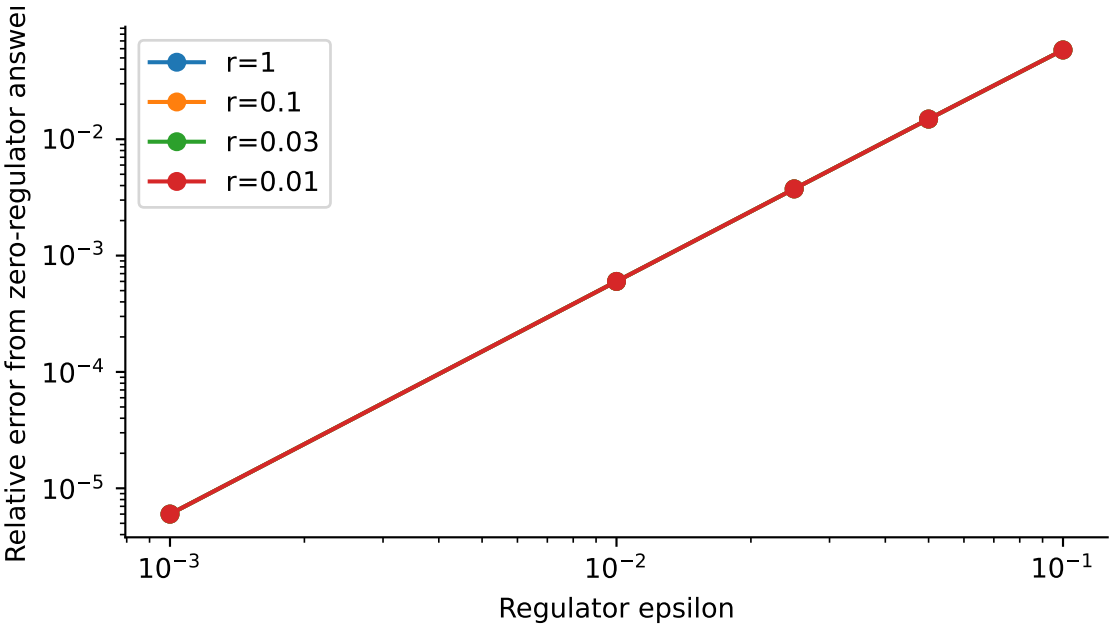


Figure 1: Regulator dependence of the independently integrated test. The curves coincide because the dimensionless relative correction depends on the regulator, not the triangle scale.

This fixes the overall sign, the combinatorial factor and the vacuum prescription before the spectator interaction is used. It is a Hamiltonian test; no unchecked derivative-interaction Legendre transform is assumed.

References

[1] N. Arkani-Hamed and J. Maldacena, *Cosmological Collider Physics*, arXiv:1503.08043.

Code and supporting data: step6 on GitHub.

Contributions (equal)

Member	Contribution
Priyansh Rawat	Heavy-clock exponent and in-in derivation.
Samarth Bhatnagar	Sign and contraction-factor checks; benchmark review.
Mayank Sinha	Independent oscillatory quadrature and regulator tests.